


Create a project in vite. Please follow the initial setup steps and delete unnecessary codes from files.

Maine components mei do files banayi → Logout.jsx
→ Person.jsx

```
import React from 'react'
import Person from './Components/Person'
import Logout from './Components/Logout'
```

```
function App() {
  return (
    <div>

      <Logout/>

    </div>
  )
}
```

```
export default App
```

→ App.jsx

```
import React from 'react'
```

```
function Person(props) {
```

```
  console.log(props);
```

```
  return (
```

```
    <div>
```

```
      <h1>Naam: {props.name}</h1>
```

```
      <h1>Umar: {props.age}</h1>
```

```
      <h1>FavColor: {props.color}</h1>
```

```
    </div>
```

```
  )
```

```
}
```

```
export default Person
```

→ Person.jsx

```
import Person from './Person'

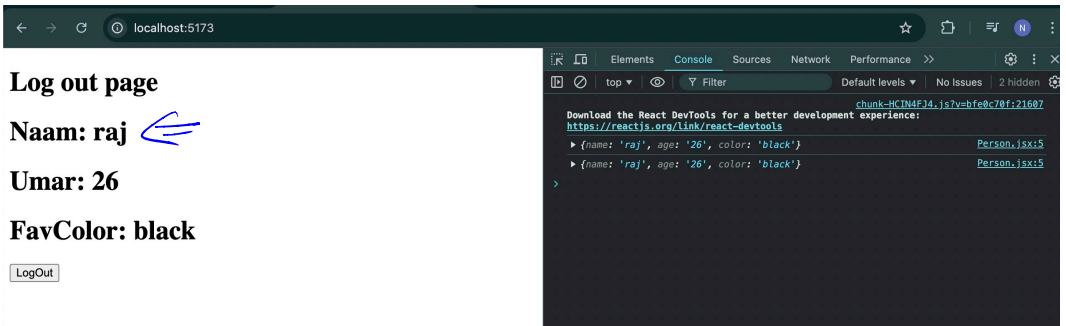
function Logout() {
  return (
    <div>
      <h1>Log out page</h1>

      <Person name="raj" age="26" color="black"/>

      {/* <Person name="saryu" age="24"
color="white"/> */}
      <button>Logout</button>
    </div>
  )
}

export default Logout
```

O/p ←



Mai chahata hu ki button par click karne par
 "Name: raj" ki jagah "Name: anonymous" hojaye dom
 mei.

I made changes in Logout.jsx code and added an
 event for onClick inside it.

Logout.js

```
import Person from './Person'

function Logout() {

  let userName = "Raj";

  function handleClick(){
    console.log(userName);
    userName= "anonymous"
    console.log(userName);
  }

  return (
    <div>
      <h1>Log out page</h1>
      <Person name= {userName} age="26" color="black"/>
      { /* <Person name="saryu" age="24" color="white"/> */ }
      <button onClick = {handleClick}>Logout</button>
    </div>
  )
}

export default Logout
```

localhost:5173

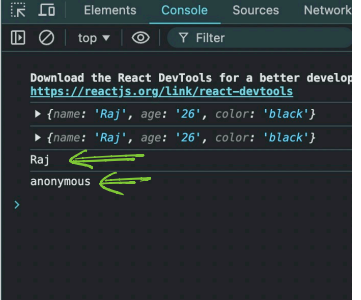
Log out page

Naam: Raj

Umar: 26

FavColor: black

Logout



Download the React DevTools for a better development experience.
<https://reactjs.org/link/react-devtools>

- ▶ {name: 'Raj', age: '26', color: 'black'}
- ▶ {name: 'Raj', age: '26', color: 'black'}

Raj

anonymous

DOM par "Raj" change nahi hua. But console mei Raj and anonymous aagya matlab userName variable ki value change ho gyi thi.

React kehata hai ki aap variable ki value change kar sakte ho but jo chiz apne sender ko di usmei change nahi aayega. Render matlab display.

Render karana padega!!

Logout.jsn mei <Person /> wale component ko re-render karana padega. But you cannot re-render directly in React; you will need to use Hook.

Hooks are pre built functions. (25:30)

useState() → Hook for re-rendering

↳ when it is called, it returns an array that has 2 elements → ① variable ka naam (username) ② setUsername()

↳ ye vo function hai jiski madad se variable ka naam change hoga.
↳ iska naam kuch bhi ho sakta hai

variable ko React ki term mei state kehte hai.

code mei useState likhte hi automatically import ho jayega.

(29:43)

Whenever there is a change in state/ variable, the component tracks the changes made and component re-renders itself.

Note: if you want to change any variable dynamically, always use useState() hook, naaki let a = 10; it won't work.

```

import { useState } from 'react'
import Person from './Person'

function Logout() {
  let [userName, setUserName] = useState("Nitin");

  function handleClick(){
    setUserName('anonymous');
  }

  return (
    <div>
      <h1>Log out page</h1>

      <Person name= {userName} age="26" color="black"/>

      <button onClick = {handleClick}>LogOut</button>

    </div>
  )
}

export default Logout

```

→ Logout.jsx

Before :-

Log out page

Naam: Nitin

Umar: 26

FavColor: black

LogOut

Download the React DevTools for a better development experience
<https://reactjs.org/link/react-devtools>

```

▶ {name: 'Nitin', age: '26', color: 'black'}
▶ {name: 'Nitin', age: '26', color: 'black'}
>

```

After :-



Log out page

Naam: anonymous

Umar: 26

FavColor: black

LogOut

Download the React DevTools for a better development experience
<https://reactjs.org/link/react-devtools>

```

▶ {name: 'Nitin', age: '26', color: 'black'}
▶ {name: 'Nitin', age: '26', color: 'black'}
▶ {name: 'anonymous', age: '26', color: 'black'}
▶ {name: 'anonymous', age: '26', color: 'black'}
>

```

execution

Code explanation :-

1. Component Initialization

- When the `<Logout>` component is rendered:
 - `useState("Nitin")` is called.
 - This initializes the `userName` state variable to `"Nitin"`. 
 - `setUserName` is the state updater function provided by `useState`.
 - React keeps track of the state of `userName` internally.

2. Initial Rendering

- React renders the JSX returned by the `Logout` component:

jsx

 Copy code

```
<div>
  <h1>Log out page</h1>
  <Person name={userName} age="26" color="black" />
  <button onClick={handleClick}>Logout</button>
</div>
```

- At this point:
 - `userName` is `"Nitin"`, so the `Person` component is called with `name="Nitin"`, `age="26"`, and `color="black"`.
 - The `button` is set up with an `onClick` event listener that calls the `handleClick` function when clicked.

l1 → yaha variable ko state variable bola gaya hai.

3. Interaction - Button Click

- When the user clicks the "LogOut" button:
 - The `handleClick` function is triggered.
 - Inside `handleClick`, `setUserName('anonymous')` is called.
 - This updates the `userName` state variable to `"anonymous"`.
 - React schedules a re-render of the `<Logout>` component.

4. Re-Rendering

- React re-renders the `<Logout>` component:
 - The updated state, `userName = "anonymous"`, is used.
 - The `Person` component is now called with `name="anonymous"`, `age="26"`, and `color="black"`.

Note:- Jaise hi state change hogi, Logout component kharf ko re-render karega. Jo chiz change hogi hai, sirf wahi change hogi, baaki same rahega.

5. Updated UI

- The browser reflects the updated state:
 - The `Person` component displays `name="anonymous"`.
 - The rest of the UI remains unchanged.

6. Key Points

- State Management:
 - The `useState` hook ensures that changes to `userName` trigger a re-render.
- Props Flow:
 - The `Person` component receives `userName` as a prop and updates its display accordingly.
- Event Handling:
 - The `handleClick` function is passed to the `button`'s `onClick` event, connecting the UI interaction with state updates.

Please note that "Nitin" is an initial value.

Doubt :-

is this setUsername an inbuilt function in react?

No, `setUserName` is not an inbuilt function in React. It is a function returned by the `useState` hook, which is a React-provided function.

Here's how it works:

1. `useState` Hook:

- When you call `useState(initialValue)`, React returns an array containing two elements:
 - State variable: This holds the current state value.
 - Updater function: A function to update the state.

javascript

Copy code

```
const [stateVariable, setStateVariable] = useState(initialValue);
```

2. In Your Code:

- When you write `let [userName, setUsername] = useState("Nitin");`:
 - `userName` is initialized to "Nitin".
 - `setUserName` is the updater function that React provides to change the value of `userName`.

3. Purpose of `setUserName`:

- `setUserName` updates the `userName` state.
- When you call `setUserName(newValue)`, React:
 - Updates the `userName` value.
 - Triggers a re-render of the component so the updated state is reflected in the UI.

Note:- "Nitin" is initial value of username state variable.

React interview question:-

Q Button par click karke hi value increment = karo.

```
import React, { useState } from 'react'

function Counter() {
  let [count, increaseCount] = useState(0);
  function increment(){
    increaseCount(count++);
  }
  return (
    <div>
      <h1>Counter</h1>
      <button onClick={increment}> increase {count}</button>
    </div>
  )
}

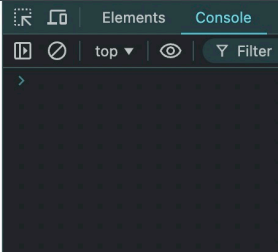
export default Counter
```

Counter

increase 15



localhost:5173



Q How to handle div nesting?

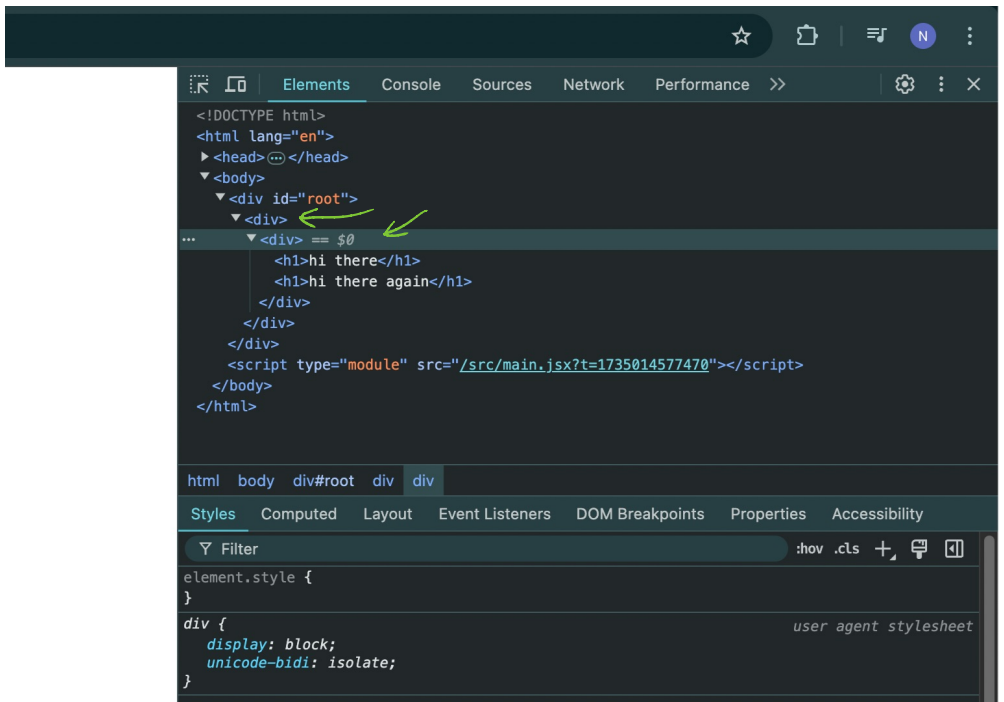
```
import React from 'react'

function Loogout() {
  return (
    <div>
      <h1>hi there</h1>
      <h1>hi there again</h1>
    </div>
  )
}

export default Loogout
```

→ Loogout.jsx

Just Bcoz mujhe do items/elements return karane the, isliye mujhe dono ko ek div mei wrap karna pada and jiske karan mere html code mei div nesting hogayi.



upr ss mei , we can see two extra divs .

<div id="root"> → this is coming from index.html

solution is → give empty tags.

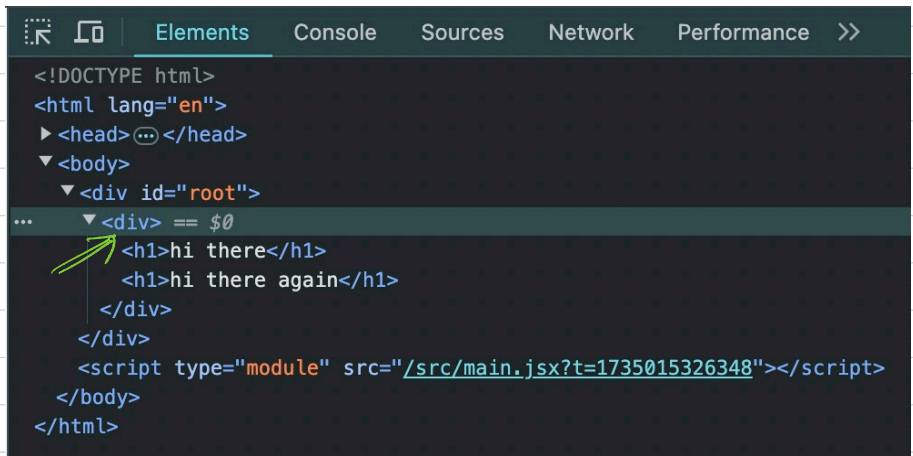
```
import React from 'react'
```

→ Logout.js

```
function Logout() {  
  return (  
    <>  
    <h1>hi there</h1>  
    <h1>hi there again</h1>  
    </>  
  )  
}
```

→ empty tag

```
export default Logout
```



```
<!DOCTYPE html>  
<html lang="en">  
  <head>...</head>  
  <body>  
    <div id="root">  
      <div> == $0  
        <h1>hi there</h1>  
        <h1>hi there again</h1>  
      </div>  
    </div>  
    <script type="module" src="/src/main.jsx?t=1735015326348"></script>  
  </body>  
</html>
```

Ab sirf ek div reh gya.

OR

you can also use <React.Fragment> instead of empty tag if you want.

<React.Fragment> </React.Fragment> in place of empty tags.

OR

you can also destructure Fragment from React and write like this :-

<Fragment> </Fragment>

```
import React, { Fragment } from 'react'

function Logout() {
  return (
    <Fragment>
      <h1>hi there</h1>
      <h1>hi there again</h1>
    </Fragment>
  )
}

export default Logout
```

This is how you can reduce div nesting.

Q return karne time sirf ek 1 parent kyu hota hai?

Bcoz of faster reconciliation

Q Re-render mtlb?

→ Dynamic changes ko DOM par display karna fir se krni by default dynamic changes khud se display nahi hote.

Re-rendering ko you can't see; you will need a React DevTools tool for it in order to see

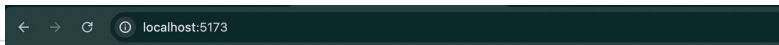
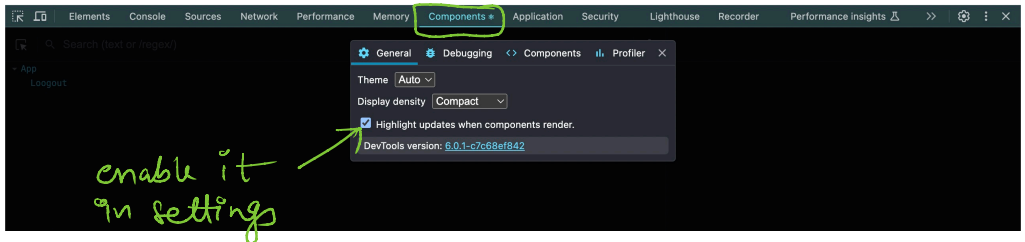
re renders.

Install Chrome extension :- React Developers tool

open "Components" in Chrome dev tool

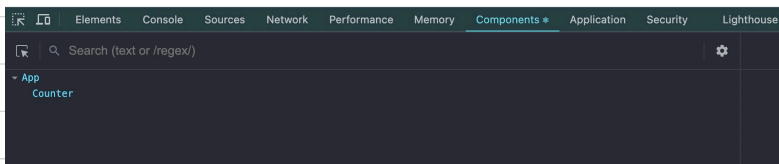
hi there

hi there again



Counter

increase 5



Ab jaise hi ye button click hoga, waise hi rendering hote hue dikhega. green green dikhega.

(1:33:00 break only)

Doubt:-

my teacher told me long ago that in react - everything is not changed/reloaded again if there is a dynamic change in something. in the code given below, there are two elements - button and h1 tag. so when i click button, does h1 tag get reloaded again or not?

```
import React, { useState } from 'react'

function Counter() {

  let [count, increaseCount] = useState(0);

  function increment(){

    increaseCount(count++);

  }

  return (
    <div>
      <h1>Counter</h1>
      <button onClick={increment}> increase {count}</button>
    </div>
  )
}

export default Counter
```

Your teacher is correct: React uses a **virtual DOM** to optimize updates. When the state changes (e.g., `count` in your example), React does **not reload everything**. Instead, it compares the virtual DOM with the actual DOM to update only the parts that have changed.

What Happens When the Button is Clicked?

1. State Update:

- The `increment` function is called when the button is clicked.
- Inside `increment`, `increaseCount(count++)` is executed:
 - `count++` increments `count` but **returns the old value** because the post-increment operator is used.
 - This may not behave as expected, as `increaseCount` might receive the wrong value.

2. Re-rendering:

- React schedules a re-render of the `<Counter>` component because the `count` state is updated.

3. Reconciliation Process:

- During the re-render, React compares the updated virtual DOM with the previous virtual DOM:
 - Static content** (e.g., `<h1>Counter</h1>`) remains unchanged because React detects no changes in this part of the DOM.
 - Dynamic content** (e.g., the `count` value displayed in the button text) is updated, and only that part of the DOM is changed.

Does the `<h1>` Tag Reload?

- No, the `<h1>` tag does not reload in the browser. React knows it hasn't changed, so it skips any updates to it.
- React updates only the parts of the DOM that are affected by the state change—in this case, the text inside the `<button>`.

→ Samarth sir kealc re-rendering ho sahi hai non-changing elements ki bhi. Chatgpt could be wrong. Sir ne visualize karke bhi dikhaya using react dev tool.

Let's use React DevTool :-

Next page ke code mei hamne do components inside one file. sir ne button par click karke dikhaya ki har chiz se sender ho sahi hai, change to change ho sahi hai. How to prevent it?

↓
green lines dikh sahi hai DOM mei

```

import React, { useState } from "react";

function Maverick() {
  let [titl, setTitl] = useState("do bhau bhau");

  function handleTodo() {
    setTitl(Math.floor(Math.random() * 10));
  }

  return (
    <div>
      <MavKeTodo title={titl} />
      <button onClick={handleTodo}>Click</button>

      <MavKeTodo title="eat treat" />
      <MavKeTodo title="hi five" />
      <MavKeTodo title="sit" />
    </div>
  );
}

function MavKeTodo(props) {
  return <h1>task: {props.title} </h1>;
}

export default Maverick;

```

→ SL
code

```

import React, { useState } from "react";

```

```

function Tida() {
  return (
    <div>
      <Head task="do roll" />
      <Head task="do siff" />
      <Head task="do bark" />
      <BadaHead task="do jump" />
    </div>
  );
}

```

```

function Head(props) {
  return <h1>Title: {props.task} </h1>;
}

```

```

function BadaHead({ task }) {
  let [t, setT] = useState(task); //do jump
  function handleClick() {
    setT(Math.floor(Math.random() * 10));
  }
  return (
    <>
      <Head task={t} />
      <button onClick={handleClick}>Click</button>
    </>
  );
}

```

```

export default Tida;

```

→ Last code

→ Maverick.jsx

⇒ Tida.jsx

→ iss code mei khaas baat ye hai ki isme sirf ek element render horaha hai balki last code mei saare element render horah the. Let's understand what's happening here!!

Note:- Webapp mei extra re-rendering hone se koi loss nahi hai but mobile app mei re-rendering ko

reduce karna chahiye but
it is best to reduce re
rendering in all devices.

SL code (second best code)
L code (best code)

L code mei maine alag function/component banaya
us element ke liye jisme dynamic change ho
saka hai.

Sir ne "do jump" ke liye "Bada Head" separately
create kiya

Fix Bada Head mei Head ko call karo.
iske tarike se you can avoid ~~extra~~ extra
rendering.

Tida mei $\xRightarrow{H}$

Head X3
Bada Head

Bada Head $\xRightarrow{H}$

Head

dynamic change
ka logic

Indirectly Head call hua, naaki directly.
variable

Jab bhi state change hoti hai tab uska
component re-render hota hai. Isliye hum iss
component (Bada Head) ko alag component
bana diya. Sirf Bada Head re-render hua,
naaki Head.

(1:59:40) Whenever a state is changed, use state variable ka component se render hoga. jals t change hoga, then sirf Body Head se render hua.

26 Dec, 2024

(2:03:50) # Memoization :-

→ ye useMemo nahi hai

→ Apne changes ko yaad rakhna

It works by tracking the props. If props are changed, iska matlab re-rendering hogi.

If props are not changed, re-rendering nahi hogi.

```
import React, { useState } from 'react'

function Memo(){
  let [t, setT] = useState('hi memo');
  function handleClick(){
    setT(Math.floor(Math.random() * 10));
  }
  return(
    <>
    <MyMemo title= {t}/>
    <button onClick= {handleClick}>click here</button>
    <MyMemo title='bye memo'/>
    <MyMemo title='good bye memo'/>
    </>
  )
}

function MyMemo({title}) {
  return (
    <div>
      <h1>{title}</h1>
    </div>
  )
}

export default Memo
```

→ Memo.jsu

Memoization kaise hoga?

Red wale function ko hum memo() mei daal denge and function mei se MyMemo text hata denge.

Upar wale code mei sabhi elements se render hoga kyunki + change hoga button click karne per isliye + ka function component Memo se render hoga. To resolve it, we will use memoization.

MyMemo wale function ko memo() mei daal do

```
import React, { memo, useState } from 'react'
```

```
function Memo(){
```

```
  let [t, setT] = useState('hi memo');
```

```
  function handleClick(){
```

```
    setT(Math.floor(Math.random() * 10));
```

```
  }
```

```
  return(
```

```
    <div>
```

```
    <MyMemo title= {t}/>
```

```
    <button onClick= {handleClick}>click here</
```

```
  button>
```

```
    <MyMemo title='bye memo' />
```

```
    <MyMemo title='good bye memo' />
```

```
    </div>
```

```
  )
```

```
}
```

```
const MyMemo = memo(function ({title}) {
```

```
  return (
```

```
    <>
```

```
    <h1>title: {title}</h1>
```

```
  </>
```

```
)
```

```
  })
```

```
export default Memo
```

Fragment &

yaha empty tag use karunga then memoization work nahi kariga kyunki re-rendering hogi. Isliye yaha sirf div tag use karoo.

yaha par koibhi tag (empty tag, fragments, div) use kar do. Sab kuch chalega.

Jo bhi chiz display hogi, use memo() mei daal do.

Note:- memo likhte hi memo destructure hojayege
header mei import wali line mei.

Memorization ke case mei, jab `<MyMemo />` ko
div parent nahi milega tab sab ke sab resender
hoge. (check L 1) isliye yaha Fragment and
empty tag use mat karo. Isliye jaha call kiga,
vaha `<div>` mei wrap karo.

